

Competition for light and water in a coupled soil-plant system



Gabriele Manoli^{a,b,*}, Cheng-Wei Huang^c, Sara Bonetti^d, Jean-Christophe Domec^{b,e},
Marco Marani^{b,f}, Gabriel Katul^{b,d}

^a Institute of Environmental Engineering, ETH Zurich, 8093 Zurich, Switzerland

^b Nicholas School of the Environment, Duke University, Durham, NC 27708, USA

^c Department of Biology, University of New Mexico, Albuquerque, NM 87131, USA

^d Pratt School of Engineering, Duke University, Durham, NC 27708, USA

^e Bordeaux Sciences Agro UMR INRA-TCEM 1220, University of Bordeaux, 33195 Gradignan, France

^f Department of Civil, Architectural and Environmental Engineering, University of Padova, via Loredan 20, 35131 Padova, Italy

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ABSTRACT

It is generally accepted that resource availability shapes the structure and function of many ecosystems. Within the soil-plant-atmosphere (SPA) system, resource availability fluctuates in space and time whereas access to resources by individuals is further impacted by plant-to-plant competition. Likewise, transport and transformation of resources within an individual plant is governed by numerous interacting biotic and abiotic processes. The work here explores the co-limitations on water losses and carbon uptake within the SPA arising from fluctuating resource availability and competition. In particular, the goal is to unfold the interplay between plant access and competition for water and light, as well as the impact of transport/redistribution processes on leaf-level carbon assimilation and water fluxes within forest stands. A framework is proposed that couples a three-dimensional representation of soil-root exchanges with a one-dimensional description of stem water flow and storage, canopy photosynthesis, and transpiration. The model links soil moisture redistribution, root water uptake, xylem water flow and storage, leaf potential and stomatal conductance as driven by supply and demand for water and carbon. The model is then used to investigate plant drought resilience of overstory-understory trees simultaneously competing for water and light. Simulation results reveal that understory-overstory interactions increase ecosystem resilience to drought (i.e. stand-level carbon assimilation rates and water fluxes can be sustained at lower root-zone soil water potentials). This resilience enhancement originates from reduced transpiration (due to shading) and hydraulic redistribution in soil supporting photosynthesis over prolonged periods of drought. In particular, the presence of different rooting systems generates localized hydraulic redistribution fluxes that sustain understory transpiration through overstory-understory interactions. Such complex SPA dynamics cannot be properly summarized by equivalent ecosystem-scale Resistor-Capacitor (RC) representation. However our results show that, with proper averaging across water flow paths, RC models can provide reasonable estimates of stand-level water and carbon fluxes during inter-storm periods.

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1. Introduction

The significance of plant water uptake and plant tolerance to drought in ecosystem management, forestry, and agriculture is rarely disputed (Fatichi et al., 2016; Jackson et al., 2000). Plant hydraulics and canopy transpiration play a major role in regulating the exchanges of mass (water and carbon) and energy at the land surface, affecting groundwater recharge, surface runoff, convective precipitation, and even eco-climate teleconnections (e.g.

Roberts, 1983; Koster et al., 2004; Jackson et al., 2005; Katul et al., 2012; Guillod et al., 2015; Bonetti et al., 2015; Manoli et al., 2016; Torngern et al., 2017; Stark et al., 2016; Vrese et al., 2016). Because of its significance, the subject of representing plants in hydrological and Earth System models (ESM) continues to draw significant research attention. Historically, the inclusion of plants in such models utilized the seminal work by Penman (1948) and Monteith (1965), where potential evapotranspiration was introduced as a sink term to the soil water budget (e.g. Feddes et al., 2001; Vrugt et al., 2001; Hopmans and Bristow, 2002), generally using a big leaf approximation (i.e. model grid elements homogeneously covered by a big leaf (Dai et al., 2004; Deardorff, 1978; Sellers et al., 1986)) and empirical water stress parameterizations

* Corresponding author at: Institute of Environmental Engineering, ETH Zurich, 8093 Zurich, Switzerland.

E-mail address: manoli@ifu.baug.ethz.ch (G. Manoli).

(Feddes et al., 2001; Volpe et al., 2013). The water stress functions are then adjusted depending on species sensitivity to drought. Other models attempt to resolve the flow processes occurring along the water pathways and across various plant organs, which are surrogated to resistances and conductances (RC) in the stem, branches and leaves (e.g. Phillips et al., 1997; Zweifel and Häsler, 2001; Zhuang et al., 2014). Such a “zeroth-order” representation still neglects many complex processes, which are fundamental for plant functioning such as non-linear pressure-volume relations within the plant, inter-facial processes occurring at the soil-root interface or leaf-atmosphere interface. As an alternative to RC systems, a 1-D “porous-medium” representation of plant hydrodynamics has been proposed to describe water transport and storage within the plant xylem (e.g. Edwards et al., 1986; Kumagai, 2001; Bohrer et al., 2005; Chuang et al., 2006; Manzoni et al., 2013). In essence, these models assume the validity of cohesion-tension theory within the xylem and represent xylem hydraulic conductivity and plant-water storage using vulnerability to embolism curves and pressure-volume relations. Such studies have successfully modeled time lags between sap flow and canopy transpiration and provided vertically-explicit description of stem and leaf water potentials (e.g. Kumagai, 2001; Chuang et al., 2006; Huang et al., 2017). However, the coupling with soil moisture was generally neglected and only recently the effect of plant water storage in regulating root-water uptake (RWU) and hydraulic redistribution (HR) in the soil have been considered (Huang et al., 2017; Mirfenderesgi et al., 2016). Other revisions have focused on the spatial branching and its network features in roots (e.g. Couvreur et al., 2012), stem and branches (e.g. Bohrer et al., 2005; Mirfenderesgi et al., 2016), display of foliage and their light interception within the canopy volume (Duursma and Mäkelä, 2007). While a structurally detailed network representation accommodates well-known scaling laws between tree structure and flow hydrodynamics (Bohrer et al., 2005), these approaches are computationally demanding (now and in the foreseeable future) to be used within ESM and remain suitable at the scale of a single plant. To begin including elementary ecosystem level processes such as simultaneous competition for light and water resources (e.g. overstory-understory) requires a novel hybrid approach that takes full advantage of all the aforementioned developments and frames the scope of this study. Hence, the goal here is to explore carbon and water exchanges within the soil-plant system when the ecosystem is comprised of multiple interacting trees competing for light and water. Specifically, we aim to disentangle the role of competing rooting systems and canopy shading in regulating soil water dynamics, plant hydraulics and canopy photosynthesis in forest biomes and test the hypothesis that such understory-overstory interactions increase ecosystem resilience to drought. With the term “ecosystem” we refer here to a forest patch composed by equally spaced overstory trees with an inter-row understory, i.e. the typical configuration of large scale pine plantations and intercropping systems in the Southeastern US (King et al., 2013; Manoli et al., 2016; Sun et al., 2010). Building upon recent advances in plant physiology (i.e. stomatal optimality (Katul et al., 2010; Manzoni et al., 2011)) and coupled soil-plant modeling (Huang et al., 2017; Manoli et al., 2014; Volpe et al., 2013), this goal is addressed by: (i) developing an innovative 3D-1D representation of the soil-plant system that links 3D subsurface soil-root processes with a primarily 1D description of above ground xylem flow and photosynthesis, (ii) investigating the effects of water and light competition among multiple and competing trees and the consequences of such competition to ecosystem resilience to drought (i.e. sustained carbon/water fluxes at low soil water potentials), (iii) evaluating the impact of tree competition on the transient nature of parameters of an equivalent RC representation, which is the likely approach to be used in large-scale ESM.

2. Mathematical model

A description of the 3D soil-root water uptake model and its coupling to a 1D representation of above-ground processes and leaf-level gas exchange is presented. The water storage within the woody tissues is explicitly modeled, the vertical distribution of leaf area density is resolved, and competition for soil water and light is accounted for. However, it is still assumed that (i) each soil-root layer is directly linked to the base of the xylem, i.e. energy losses as water flows within the rooting system are negligible compared to energy losses at the soil-root interface (Siqueira et al., 2008), (ii) nutrient availability is not a limiting factor at the hydrological time scales of interest (Manoli et al., 2014), and (iii) plant growth occurs at time scales much longer than the simulation period so that no structural changes in the rooting system, above ground biomass, and leaf area occur (Manoli et al., 2014; 2015).

A “beam” model is used for the above ground compartment (see Fig. 1) in accordance with the cantilever beam representation of trees employed in computational fluid dynamics (Dupont et al., 2015; Pivato et al., 2014). The branching structure of the canopy is simplified to a multi-layer canopy so that the stem water potential at each layer represents an average stem-branches potential regulating stem-leaf flow and xylem storage. The plant model is kept intentionally simple to reduce the computational costs given that much of the light environment heterogeneity to be considered here is in the vertical direction.

A list of symbols and abbreviations used in the text and equations below is provided in Table 1.

2.1. Soil-root exchanges

Richards' equation is used to describe soil moisture dynamics and RWU in a three-dimensional porous medium (domain Ω in Fig. 1), as implemented in the CATHY model (Camporese et al., 2010; Manoli et al., 2014) and is given by:

$$S_s S_w(\psi) \frac{\partial \psi}{\partial t} + n \frac{\partial S_w(\psi)}{\partial t} = \vec{\nabla} \cdot [\mathbf{K}_s \mathbf{K}_r(\psi) (\vec{\nabla} \psi + \eta_z)] - q_r \quad \text{in } \Omega \in \mathbb{R}^3, \quad (1)$$

where S_s is the elastic storage term [m^{-1}], S_w is water saturation [-], ψ is the soil water potential [m], t is time [s], n is the porosity [-], $\mathbf{K}_s$ is the saturated hydraulic conductivity [m s^{-1}] tensor, K_r is the relative hydraulic conductivity [-], $\eta_z = (0, 0, 1)^T$ is the gravitational potential energy gradient with z , the vertical coordinate, directed upward. The flux $q_r = q_r(\psi, x, y, z, t, \psi_{sb,j})$ is a macroscopic source/sink term [s^{-1}] through which soil water dynamics are coupled with each root-plant system j in the domain Ω via the water potential at the base of each stem $\psi_{sb,j} = \psi_p(z_{sb,j})$, ψ_p is the xylem water potential and $z_{sb,j}$ the elevation of the stem base. Single plants are defined by a surface grid node j (with j between 1 and the number of plants) and the three-dimensional RWU flux (per unit soil volume) is given by Manoli et al. (2014, 2015):

$$q_{r,j} = -g_{sr,j} [(\psi_{sb,j} + z_{sb,j}) - (\psi + z)] a_R, \quad (2)$$

where $g_{sr,j}$ is the total soil-to-root conductance [s^{-1}] and $a_R = 2\pi r B(x, y, z)$ is the total root surface area per unit soil volume [$\text{m}_{\text{root}}^2 \text{m}_{\text{soil}}^{-3}$], r is the effective root radius and $B(x, y, z)$ is the root length density [$\text{m}_{\text{root}} \text{m}_{\text{soil}}^{-3}$]. The net root water uptake (RWU_{net}) by plant j is given by:

$$Q_j = \int_0^{V_{r,j}} q_{r,j} dV_{r,j}. \quad (3)$$

where $V_{r,j}$ is the volume of soil occupied by the root biomass of plant j . During night $q_{r,j}$ can assume both positive (uptake) and negative (redistribution) values within the root zone. Therefore, the

Table 1
Notation and abbreviations.

Symbol	Description	Units
a	Relative diffusivity of water vapor with respect to CO ₂	–
$a_{1,j}$	Photosynthetic parameter of plant j	$\mu\text{mol m}^{-2} \text{s}^{-1}$
$a_{2,j}$	Photosynthetic parameter of plant j	$\mu\text{mol mol}^{-1}$
a_p	Tapering coefficient for stem area	–
a_R	Root surface area per unit volume	m^{-1}
$A_{b,j}$	Branch cross-sectional area of plant j	m^2
A_c	Total projected canopy area	m^2
$A_{c,j}$	Total projected canopy area of plant j	m^2
A_e	Surface area of the Ω domain	m^2
$A_{p,j}$	Sapwood cross-sectional area of plant j	m^2
$A_{p,max,j}$	Cross-sectional area of the stem base of plant j	m^2
$B(x, y, z)$	Root length density	m m^{-3}
β	Parameter for the marginal water use efficiency	m^{-1}
c_a	Ambient CO ₂ concentration	mmol mol^{-1}
c_a^0	Reference CO ₂ concentration	mmol mol^{-1}
c_{cp}	CO ₂ Compensation point (at 25°C)	mmol mol^{-1}
$c_{ic,j}$	Inter-cellular CO ₂ concentration	mmol mol^{-1}
c_{K_p}	Parameter for the xylem vulnerability to embolism curve	–
c_{g_0}	Parameter for cuticular conductance	–
$c_{o,a}$	O ₂ concentration in the atmosphere	mmol mol^{-1}
$C_{p,j}$	Capacitance of sapwood for plant j	m^{-1}
$C_{p,eq}$	Ecosystem-scale capacitance of sapwood	m^{-1}
d_{K_p}	Parameter for the xylem vulnerability to embolism curve	m
d_{g_0}	Parameter for cuticular conductance	m
Δz	1D grid spacing	m
ϵ_w	Conversion coefficient ($=M_w/\rho_w$)	$\text{m}^3 \text{mol}^{-1}$
f_c	Leaf-scale carbon assimilation flux	$\mu\text{mol m}^2 \text{s}^{-1}$
f_e	Leaf-scale transpiration flux	$\text{m}^3 \text{s}^{-1}$
γ_j	Apparent quantum yield of plant j	–
Γ_j	1D domain for plant j	–
$g_{b,j}$	Xylem conductance of branches of plant j	s^{-1}
$g_{p,eq}$	Ecosystem-scale xylem conductance	s^{-1}
$g_{sr,j}$	Total soil-to-root conductance	s^{-1}
$g_{sp,eq}$	Ecosystem-scale soil-plant conductance	s^{-1}
$g_{st,j}$	Stomatal conductance of plant j	$\text{mol m}^{-2} \text{s}^{-1}$
$g_{st,eq}$	Ecosystem-scale stomatal conductance	$\text{mol m}^{-2} \text{s}^{-1}$
$g_{o,j}$	Cuticular conductance to water vapour of plant j	$\text{mol m}^{-2} \text{s}^{-1}$
$g_{o,max}$	Maximum cuticular conductance	$\text{mol m}^{-2} \text{s}^{-1}$
$h_{p,j}$	Height of plant j	m
HR	Hydraulic redistribution	$\text{m}^3 \text{s}^{-1}$
$K_{c,j}$	Michaelis constant for CO ₂ fixation of plant j (at 25°C)	$\mu\text{mol mol}^{-1}$
$K_{o,j}$	Michaelis constant for O ₂ fixation of plant j (at 25°C)	mmol mol^{-1}
$K_{p,j}$	Xylem hydraulic conductivity of plant j	m s^{-1}
$K_{p,max}$	Maximum xylem hydraulic conductivity	m s^{-1}
K_r	Relative hydraulic conductivity	–
$\mathbf{K_s}$	Saturated hydraulic conductivity tensor	m s^{-1}
$l_{b,j}$	Branches length of plant j	m
LAI	Leaf area index	$\text{m}^2 \text{m}^{-2}$
LAD_j	Leaf area density of plant j ($= \text{LAI}$ per unit length)	$\text{m}^2 \text{m}^{-3}$
LAD_{ou}	Ecosystem (understory and overstory) leaf area density	$\text{m}^2 \text{m}^{-3}$
λ_j	Marginal water use efficiency of plant j	$\mu\text{mol mol}^{-1}$
$\lambda_{max,j}$	Maximum marginal water use efficiency of plant j	$\mu\text{mol mol}^{-1}$
Λ_j	Hamiltonian for plant j	$\mu\text{mol m}^{-2} \text{s}^{-1}$
M_w	Molecular weight of water	g mol^{-1}
n	Porosity	–
ODE	Ordinary differential equation	–
Ω	3D Soil domain	–
$p_{\theta,j}$	Empirical coefficient for xylem water content of plant j	–
PAR	Photosynthetically Active Radiation	W m^{-2}
PDE	Partial differential equation	–
ψ	Soil water potential	m
$\psi_{0,j}$	Empirical coefficient for xylem water content of plant j	m
$\psi_{L,j}$	Leaf water potential of plant j	m
$\bar{\psi}_{L,j}$	Average $\psi_{L,j}$ over the past 24 hours	m
$\psi_{p,j}$	Xylem (sapwood) water potential of plant j	m
$\psi_{sb,j}$	Water potential at the stem base of plant j	m
$q_{p,j}$	Sap flux of plant j	$\text{m}^3 \text{s}^{-1}$
q_r	Source/sink term ($= \sum_j q_{r,j}$)	s^{-1}
$q_{r,j}$	RWU flux by plant j per unit volume	s^{-1}
Q_j	Net RWU by plant j ($= \text{RWU}_{net,j}$)	$\text{m}^3 \text{s}^{-1}$
Q_j^+	Total RWU by plant j	$\text{m}^3 \text{s}^{-1}$
Q_j^-	HR flux by plant j	$\text{m}^3 \text{s}^{-1}$
r	Effective root radius	m
RWU	Root water uptake	$\text{m}^3 \text{s}^{-1}$

(continued on next page)

(Bohrer et al., 2005; Chuang et al., 2006), the xylem water content is defined as:

$$\theta_{p,j}(\psi_{p,j}) = \theta_{p,sat,j} \left(\frac{\psi_{0,j}}{\psi_{0,j} - \psi_{p,j}} \right)^{p_{\theta,j}}, \quad (6)$$

where $\theta_{p,sat,j}$ [kg m⁻³] represents the xylem water content near saturation and $p_{\theta,j}$ and $\psi_{0,j}$ are empirical coefficients (Bohrer et al., 2005; Chuang et al., 2006). From Eq. (6), an expression for the capacitance $C_{p,j}$ can be derived and is given as:

$$C_{p,j}(\psi_{p,j}) = \theta_{p,sat,j} \frac{p_j}{\psi_{0,j}} \left(\frac{\psi_{0,j} - \psi_{p,j}}{\psi_{0,j}} \right)^{-(p_{\theta,j}+1)}. \quad (7)$$

The one-dimensional canopy is characterized by a leaf area density $LAD(z)$ at height z (i.e. leaf area per unit length, [m² m⁻³]) that abides by the normalizing property $\int_0^{h_c} LAD(z) dz = LAI$, where LAI is the leaf area index and h_c is the canopy height. The LAD sets the sites for subsequent phase transition and loss of water from the tree to the atmosphere via

$$T_j(z) = \begin{cases} g_{b,j} A_{b,j}(z) (\psi_{p,j} - \psi_{L,j}) & \text{if } LAD_j(z) > 0, \\ 0 & \text{if } LAD_j(z) = 0, \end{cases} \quad (8)$$

where $A_{b,j}$ is the branch cross-sectional area, $\psi_{L,j}$ is the leaf water potential and $g_{b,j} = K_{p,j}/l_{b,j}$ [s⁻¹] is the xylem conductance of a branch with length $l_{b,j}$ (Kumagai, 2001).

Eq. (4) is solved in the domain $\Gamma_j = \{z \mid z \in [z_{sp,j}, h_{p,j}]\}$ and the coupling between the below- and above-ground compartments (i.e. Eqs. (1) and (4)) is accomplished using the mass continuity resulting in the boundary condition $q_{p,j}(z_{sb,j}) = Q_j$ at the stem base (being $Q_j = RWU_{net,j}$).

2.3. Canopy transpiration and photosynthesis

The solution of the coupled systems of Eqs. (1)–(4) requires the calculation of leaf water potential $\psi_{L,j}$. The mass continuity of the branch-leaf system is now used to link $\psi_{L,j}$ to key leaf-level processes. For each above-ground layer, the water loss is (Chuang et al., 2006):

$$T_j(z) = f_{e,j}(z) LAD_j(z) \Delta z A_{c,j}, \quad (9)$$

where $A_{c,j}$ is the total projected canopy area of plant j and leaf-scale transpiration $f_e(z)$ is given by Manoli et al. (2014)

$$f_{e,j}(z) = \begin{cases} [ag_{st,j}(\lambda_j) + g_{0,j}] VPD \epsilon_w & \text{if } \Lambda_j(z) > 0, \\ g_{0,j} VPD \epsilon_w & \text{if } \Lambda_j(z) < 0. \end{cases} \quad (10)$$

Here, $\Lambda_j(z)$ is the Hamiltonian of the stomatal optimization theory (described later) and measures the imbalance between carbon gains and water losses (in carbon units). The leaf water vapor flux f_e is driven by mean air temperature, relative humidity, CO₂ concentration and photosynthetically active radiation via their effects on stomatal conductance $g_{st,j}^*(\lambda_j) = ag_{st,j}(\lambda_j) + g_{0,j}$ [mol m⁻² s⁻¹] and vapour pressure deficit VPD [mol mol⁻¹] (being ϵ_w a conversion coefficient (Manoli et al., 2014)). Here, $g_{st,j}$ is the stomatal conductance to CO₂, $a=1.6$ is the relative diffusivity of water vapor with respect to CO₂, and $g_{0,j}$ is the cuticular conductance to water vapour (accounting also for imperfect stomatal closure) and is thought to be responsible for nocturnal water losses. The conductance $g_{st,j}$ is determined according to Katul et al. (2010) and accounts for (i) Fickian mass transfer of CO₂ ($f_{c,j}$) and H₂O ($f_{e,j}$) through stomata, (ii) a biochemical demand function described by a leaf-level photosynthesis model (Farquhar et al., 1980) and (iii) an optimal leaf water use strategy that maximizes carbon

gain for a given water loss and links the marginal water use efficiency $\lambda_j = \lambda_{max,j} \frac{c_a}{c_a^*} (\beta \cdot \bar{\psi}_{L,j})$ to a time-averaged leaf water potential $\bar{\psi}_{L,j}$ (Manzoni et al., 2011) (being c_a and c_a^* the ambient and reference CO₂ concentration). The value of $\bar{\psi}_{L,j}(z)$ is the average of $\psi_{L,j}(z)$ over the past 24 h period, which is assumed to be the minimum time scale over which the marginal water use efficiency (analogous to the currency linking leaf water and carbon economies together) senses water stress. During water stress conditions, $\bar{\psi}_{L,j}$ decreases and the value of λ_j increases leading to a decreased leaf transpiration. Complete stomatal closure is assumed to occur when the leaf water cost in carbon units exceeds the carbon gain, i.e. when $\Lambda_j(z) = f_{c,j}(z) - \lambda_j(z) \cdot ag_{st,j} VPD \epsilon_w < 0$. The cuticular conductance $g_{0,j}$ is a water loss over which plants have no control (generally assumed constant (Huang et al., 2017; Manoli et al., 2014)) and is therefore not included in the optimality hypothesis, leading to the residual flux in Eq. (10). However, residual leaf transpiration is likely to decline under dry conditions and, given the lack of observations on the dependence of $g_{0,j}$ on leaf water potential, an empirical “vulnerability curve” is employed, similar to the function describing cavitation in the xylem (Daly et al., 2004; Volpe et al., 2013). This function is expressed as $g_{0,j} = g_{0,max} \exp \left[-(-\bar{\psi}_{L,j}/d_{g_0})^{c_{g_0}} \right]$.

Combining Eqs. (8) and (9), the canopy leaf water potential (where $LAD_j(z) > 0$) is given by

$$\psi_{L,j} = \psi_{p,j} - \frac{f_{e,j}(z) LAD_j(z) \Delta z A_{c,j}}{g_{b,j} A_{b,j}(z)}. \quad (11)$$

At each active canopy layer (i.e. where $LAD_j(z) > 0$), leaf photosynthesis is determined from Farquhar et al. (1980)

$$f_{c,j}(z) = \begin{cases} \frac{a_{1,j}(z)}{a_{2,j}(z) + c_{ic,j}(z)} (c_{ic,j}(z) - c_{cp}) & \text{if } \Lambda_j(z) > 0, \\ 0 & \text{if } \Lambda_j(z) < 0, \end{cases} \quad (12)$$

where $a_{1,j}$ and $a_{2,j}$ are photosynthetic parameters that depend on whether RuBP regeneration (usually but not necessarily corresponds to light limitations) or Rubisco (associated with air temperature limitations) governs photosynthesis (Manoli et al., 2014; Volpe et al., 2013). When light is limiting $a_{1,j} = \gamma_j PAR_j(z)$ and $a_{2,j} = 2c_{cp,j}$, being γ_j the apparent quantum yield [-] and $PAR_j(z)$ the photosynthetic active radiation at canopy height z , while in the case of Rubisco limitation $a_{1,j} = V_{c,max,j}$ and $a_{2,j} = K_{c,j}(1 + C_{o,a}/K_{o,j})$, where $V_{c,max,j}$ is the maximum carboxylation capacity, $K_{c,j}$ and $K_{o,j}$ are the Michaelis constants for CO₂ and O₂ fixation, respectively, and $C_{o,a}$ is the oxygen concentration in the atmosphere (Manoli et al., 2014; Volpe et al., 2013).

The vertical distribution of light as needed in the calculations of $a_{1,j}$ and $a_{2,j}$ is determined using a standard light attenuation model with an extinction coefficient accounting for $LAD_j(z)$, leaf angle distribution and solar zenith angle (Campbell and Norman, 1998; Novick et al., 2004). In the case of light competition, an ecosystem-scale overstory-understory leaf area density, LAD_{ou} , is attributed to each shaded tree and used in the calculation of the extinction coefficient (see Fig. 1). The canopy is assumed to be a homogeneous isotropic medium characterized by leaf area distribution (LAD_{ou}) defined as the weighted average of tree-level LAD_j :

$$LAD_{ou}(z) = \frac{1}{A_c} \sum_j LAD_j(z) \cdot A_{c,j}, \quad (13)$$

where A_c is the total area covered by trees within the model domain (accounting for canopies overlapping).

3. Numerical simulations

3.1. Simulations setup

Using a setup similar to the one previously considered (Manoli et al., 2014), the model is first tested using meteorological data from a Loblolly pine plantation in North Carolina, USA (US-NC2 in the Ameriflux database). A description of the study area (climate, soil type, etc.) as well as soil hydraulic characteristics and plant physiological parameters for Loblolly pines can be found elsewhere (Domec et al., 2012; Manoli et al., 2014; Sun et al., 2010). Soil-root parameters are based on Manoli et al. (2014) where model simulations were calibrated using soil moisture data. For simplicity, the same set of physiological and hydraulic parameters is used for both understory and overstory (i.e. young vs mature trees) as inter-species competition, shade tolerance versus intolerance differences, or leaf nitrogen gradients are beyond the scope here. However, a discussion on this assumption is presented in Section 5. Dry-down simulations are run using daily periodic atmospheric variables (i.e. air temperature, air humidity, PAR and incoming shortwave radiation) measured on a typical clear sky summer day (July 14, 2007) every 30 min.

The model is first calibrated for a single tree using a 1D soil domain (1D-1D simulation). Calibration is performed using field observations of water fluxes (from sap flow and eddy covariance measurements) and leaf/branch water potentials. Stem sap flux measurements were made on six overstory trees at the height of 1.4 m above ground using heat dissipation probes (see Domec et al., 2009 for details). The sensor signal was converted to sap flux density and scaled to tree-scale transpiration using sapwood area and stand tree density (Granier and Gross, 1987). Mean stand transpiration, calculated as the average among the six trees, is considered here for calibration. Leaf (ψ_L) and branch (ψ_p) water potentials were measured on four different trees and on three needles per tree located in the upper crown section (13 m above ground) using a pressure chamber (PMS Instruments, Albany, OR, USA). Additional details on field monitoring can be found in Domec et al. (2009); 2012; Manoli et al. (2014); Sun et al. (2010). Both overstory and understory configurations are tested in 1D-1D mode, and the effect of light attenuation on understory vegetation is evaluated. The full 3D-1D soil-plant simulations (accounting for two overstory and one understory trees, Fig. 2d) are then conducted for three different scenarios (Fig. 2a–c): (i) non-competing understory-overstory, (ii) competition for water (i.e. overlapping overstory/understory root systems, overstory light attenuation neglected), and (iii) competition for both water and light (i.e. overlapping roots and overstory light attenuation).

The main model parameters are listed in Table 2. Additional information on model equations and parameters can be found elsewhere (Huang et al., 2017; Manoli et al., 2014; 2015).

3.2. Equivalent resistor-capacitor model

The 3D-1D simulations are employed to estimate the effective parameters of a single effective resistor-capacitor (RC) model (Fig. 2e). Considering two plant compartments (woody tissues and leaves) connected through a series of conductances (Bohrer et al., 2005; Manzoni et al., 2014) and neglecting storage in the leaves, the ecosystem-scale RWU and transpiration fluxes are defined as:

$$\langle Q_j \rangle_e = g_{sp,eq} \cdot (\langle \psi \rangle_s - \langle \psi_{p,j} \rangle_e) \cdot A_e, \quad (14)$$

$$\langle T_j \rangle_e = g_{p,eq} \cdot (\langle \psi_{p,j} \rangle_e - \langle \psi_{L,j} \rangle_e) \cdot \langle A_{p,j} \rangle_e, \quad (15)$$

$$= g_{st,eq} \cdot VPD \cdot \langle LAI_j \rangle_e \langle A_{c,j} \rangle_e \cdot \epsilon_w, \quad (16)$$

Table 2

Model parameters for overstory (Γ_1 , Γ_3) and understory (Γ_2) trees. Additional model parameters can be found in Manoli et al. (2014); 2015).

Parameter	Unit	Overstory	Understory
Trees	–	Γ_1 , Γ_3	Γ_2
Root depth ^a	m	2.0	0.6
Canopy height	m	17.0	6.0
A_c	m ²	12.25	9.0
A_p	m ²	0.013	0.01
LAI	m ² m ⁻²	5.52	3.19
a_p	–	0.75	0.75
l_b	m	1.75	1.5
A_b/A_p	–	0.15	0.15
$K_{p,max}$	m s ⁻¹	2.5×10^{-5}	2.5×10^{-5}
c_{K_p}	–	3.5	3.5
d_{K_p}	m	480	480
c_{g_0}	–	3.5	3.5
d_{g_0}	m	300	300
$\theta_{p,sat}$	m ³ m ⁻³	0.57	0.57
p_θ	–	100	100
λ_{max}	μ mol mol ⁻¹	664	664
β	μ m ⁻¹	–0.01	–0.01
γ	–	0.015	0.015

^a Root density is assumed to decrease linearly to zero along the lateral direction (being zero at 1.5 and 1.0 m from the trunk for overstory and understory trees, respectively).

where A_e is the total surface area of the 3D model domain, and $g_{sp,eq}$, $g_{p,eq}$, $g_{st,eq}$ are the ecosystem-scale soil-plant, xylem and stomatal conductances, respectively. The operators $\langle \cdot \rangle_s$ and $\langle \cdot \rangle_e$ indicate root-zone and ecosystem scale spatial averaging (i.e. average among multiple trees of tree level volume-averaged quantities). The values of the equivalent conductances are thus estimated by averaging the 3D-1D simulation results as follows:

$$g_{sp,eq} = \frac{\langle Q_j \rangle_e}{(\langle \psi \rangle_s - \langle \psi_{p,j} \rangle_e) \cdot A_e}, \quad (17)$$

$$g_{p,eq} = \frac{\langle T_j \rangle_e}{(\langle \psi_{p,j} \rangle_e - \langle \psi_{L,j} \rangle_e) \cdot \langle A_{p,j} \rangle_e}, \quad (18)$$

$$g_{st,eq} = \frac{\langle T_j \rangle_e}{VPD \cdot \langle LAI_j \rangle_e \langle A_{c,j} \rangle_e \cdot \epsilon_w}. \quad (19)$$

The equivalent capacitance is calculated as Bohrer et al. (2005),

$$C_{p,eq} = \frac{1}{V_{p,eq}} \frac{dS_{p,eq}}{d\langle \psi_{p,j} \rangle_e}, \quad (20)$$

where $V_{p,eq}$ is the equivalent stem volume (i.e. the sum of stem volumes of the different trees) and $S_{p,eq} = \sum_j \int_0^{z_{top,j}} \theta_j(z) A_{p,j}(z) dz$ is the total water storage of the trees in the plot.

It is to be noted that the mathematical structure of the porous media and RC approaches appreciably differ. The porous media approach requires solving a system of non-linear diffusive PDEs in space and time whereas the RC approach requires solving one ODE in time (Chuang et al., 2006).

4. Results and discussion

4.1. Water potentials and fluxes within the SPA system (1D-1D simulations)

Model results are first compared against field measurements (Fig. 3). The observed delay between RWU and transpiration due to water storage within the plant tissues (illustrated by the lag between stem sapflow and above-canopy EC measurements in Fig. 3a)

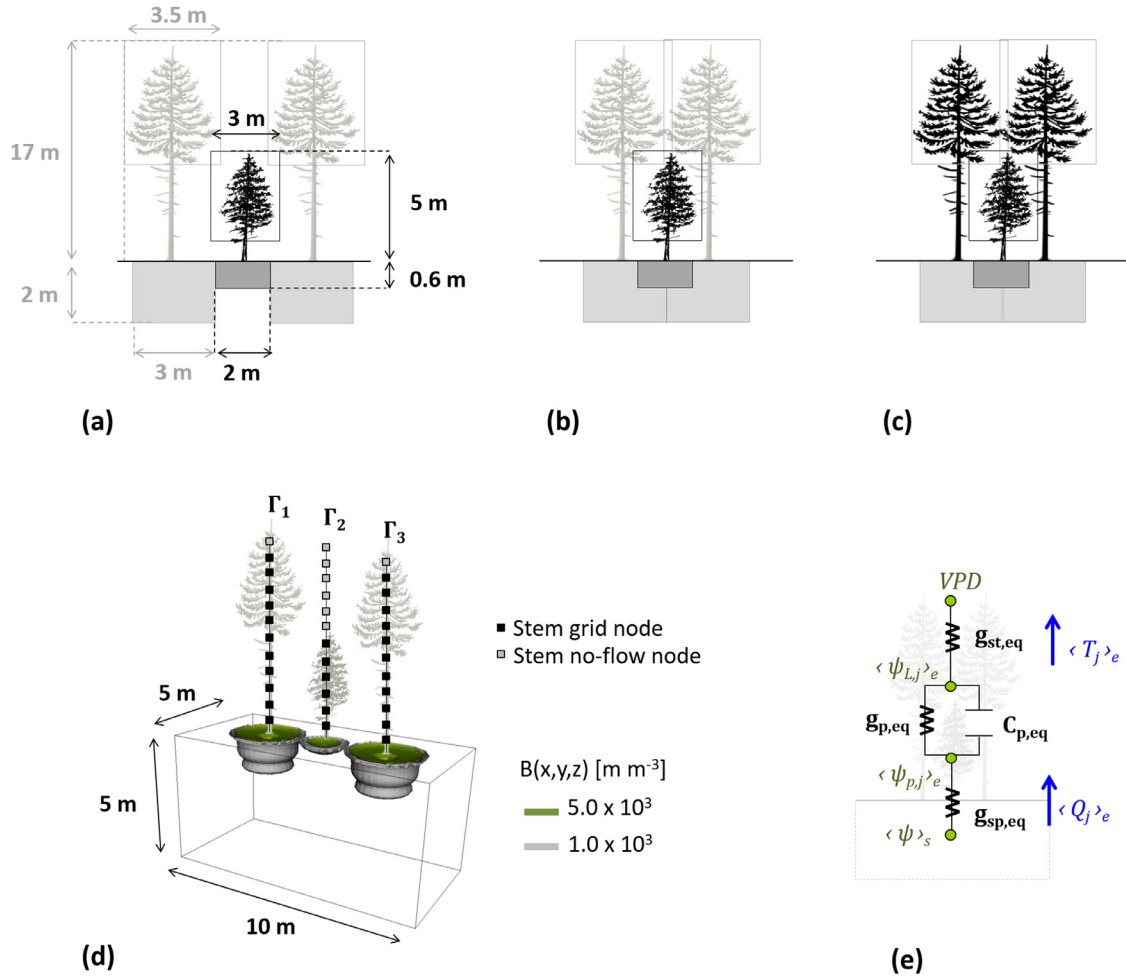


Fig. 2. Simulations setup: tree characteristics and spacing for (a) no-competition, (b) water competition, and (c) light and water competition scenarios (black/gray trees indicate shading/non-shading conditions). The 3D-1D soil-plant model domain showing root density distribution B [m m^{-3}] (gray/green contour lines) and plant stem grid nodes for trees $\Gamma_{j=1,2,3}$ is illustrated in (d). Note that the vertical root distribution of the overstory trees is based on field observations (Domec et al., 2012; Manoli et al., 2014). Along the lateral direction the root biomass is assumed to decrease linearly to zero. The equivalent RC model accounting for equivalent soil-plant, xylem and stomatal conductances ($g_{sp,eq}$, $g_{p,eq}$, $g_{st,eq}$, respectively) and ecosystem-averaged water potentials ($\langle \psi_s \rangle_e$, $\langle \psi_{p,j} \rangle_e$, $\langle \psi_{L,j} \rangle_e$) and fluxes ($\langle Q_j \rangle_e$, $\langle T_j \rangle_e$), is also shown in (e). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

is small but reasonably captured by the model. Also the water potential gradient between stem and leaves is within expected bounds set by the measurements, and the modeled diurnal cycle of water fluxes and potentials within a tree (Fig. 3b) reasonably matches the field measurements. Discrepancies between modeled and observed water potentials are attributable to the fact that ψ_L and ψ_p were measured once a day for several days of the year and for different trees (see Section 3). As discussed in Manoli et al. (2014) measurements show a large tree-to-tree variability and model simulations provide a description of the mean stand behavior only. The dry-down case in 1D-1D mode (considering an overstory tree) provides an illustration of the development of water stress conditions (Fig. 3c and d). As dry-down progresses, the canopy transpiration reduces the root-zone soil water potential which, in turn, results in a decrease in stem and leaf water potentials and subsequently transpiration itself.

The vertical distribution of fluxes along the soil-plant profile is illustrated in Fig. 4. As water stress increases, the xylem tissues become drier and drier, starting from the canopy top, where transpiration is first suppressed. With progressive xylem tissue losing hydraulic capacity (i.e. more and more cavitation), water flow within the whole SPA continuum is suppressed. Interestingly, model simulations suggest that, during summer, transpiration can be sup-

pressed also under well watered conditions (i.e. at the beginning of simulation) in the upper canopy layers (i.e. where PAR is not expected to be a limiting factor). These results suggest that the most peripheral parts of the plant are the most vulnerable and set the critical pressure on the hydraulic system, thus supporting the hydraulic segmentation hypothesis proposed by Zimmermann (1978). The modeled stomatal closure at the canopy top during mid-day conditions (Fig. 4b) is the result of low leaf water potential (high λ) and a shift from light to Rubisco limitations on photosynthesis that decreases the carbon gain f_c and, according to the optimality hypothesis, suppresses transpiration (Fig. 5). Before reaching unfavorable cost-gain conditions (negative Δ), carbon assimilation (f_c) is higher at the canopy top while the water use efficiency ($\text{WUE} = f_c/f_e$) has a minimum in the mid-low canopy (Fig. 5a–c). As shown in Fig. 4, above- and below-ground water fluxes are coordinated in such a way that they are first suppressed at the canopy top (distal part of the plant) and then in the deeper soil layers. During night-time, the water potential gradients between roots and soil generate a reverse flow of water (i.e. HR) that acts to wet the driest soil layers (Fig. 4i–l). However, nocturnal RWU and HR act as competing water fluxes at night (Huang et al., 2017). Under well-watered conditions, HR consists of 13% of daily transpiration and 60–80% of the total nocturnal RWU (Q^+) but, as the soil

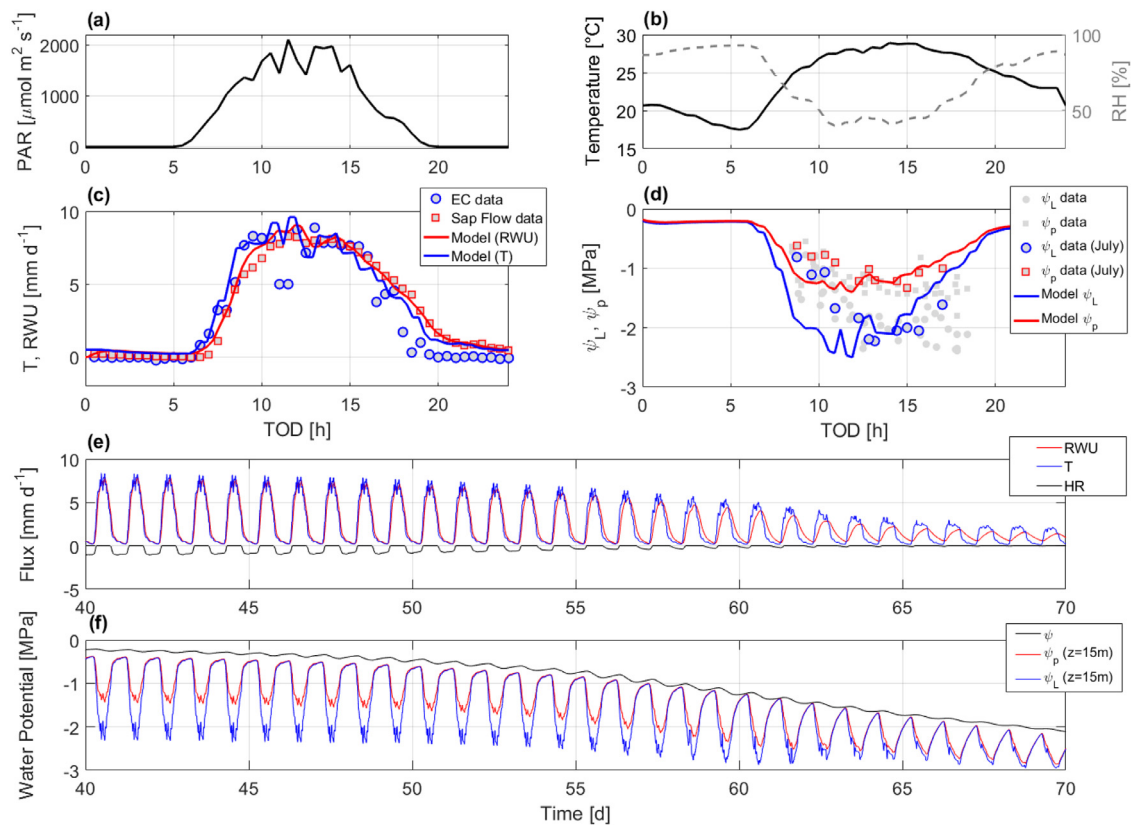


Fig. 3. 1D-1D Model simulations: meteorological forcing (a-b) and model results (c-f). Daily temporal evolution (time of day, TOD) of (a) photosynthetic active radiation PAR, and (b) temperature (solid line) and relative humidity RH (dashed line), measured on July 14, 2007 at the US-NC2 site (Domec et al., 2012; Manoli et al., 2014; Sun et al., 2010). Observed and modeled (c) RWU and transpiration (T) fluxes, (d) leaf and xylem water potential (ψ_L and ψ_p at $z = 15$ m, respectively). All available water potential data are shown (gray symbols) but, given that the model is run using the meteorological forcing recorded on July 14, (a-b), measurements collected in July are highlighted (red/blue symbols). The temporal dynamics of (e) RWU, transpiration, HR fluxes and (f) soil (at $z = 0$ m), leaf and xylem water potential during an extended dry-down period are also shown. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

dries, HR declines and the partitioning of Q^+ between RWU_{net} and HR shifts in favor of RWU_{net} (Fig. 6b). These results confirm that under severe water stress, increased nocturnal RWU and storage of water within the plant tissues is promoted rather than HR so as to maintain favorable plant water potentials (Huang et al., 2017).

To evaluate the effect of light attenuation (without considering other confounding effects, such as competition for soil water by multiple trees), a dry down simulation for an understory tree subject to different light regimes is illustrated in Fig. 7. Due to their limited rooting depth, understory trees attain water stress conditions much sooner when compared to the overstory if subjected to direct light (Fig. 7b and f). However, when light attenuation by a hypothetical overstory canopy is considered, understory transpiration is drastically reduced. This transpiration reduction thus enhances understory resilience to drought (Fig. 7d and h). In this “model world”, overstory-understory interactions may support some ecosystem functioning, prompting interest in how multiple trees function. The combined effect of light attenuation and soil water depletion by overstory-understory trees is therefore analyzed next.

4.2. Overstory-understory competition for light and water (3D-1D simulations)

The results of 3D-1D simulations are illustrated in Figs. 8 and 9. The competition for water between understory and overstory trees reduces understory transpiration. This reduction can be explained by the shallow rooting system where the available soil water is rapidly dried out due to overstory RWU. When light attenuation is

further considered, understory transpiration is reduced at an even lower rate but is maintained for a longer period of time, thereby enhancing understory resilience to drought (Fig. 10a and b).

Model results show that the decline in understory transpiration is timed to overstory HR, supporting the hypothesis that shallow tree roots may have indirect access to deep soil water through overstory HR (Quijano et al., 2012; Robinson et al., 2012; Scott et al., 2008; Yu and D’Odorico, 2014). It is to be noted that simulated HR by the overstory represents about 10–30% of its transpiration (Fig. 8i–p), which is in accordance with literature values (Neumann and Cardon, 2012) and comparable with simulated transpiration rates by the understory. The modeled spatial distribution of fluxes in the soil reveal that HR is not simply a vertical redistribution of water phenomenon (as necessarily assumed in all one-dimensional SPA models) but planar redistribution also occurs (Fig. 9). The location of HR fluxes is controlled by the presence of low water potential regions that develop in the soil due to water potential/soil moisture inhomogeneities (Robinson et al., 2012). Such inhomogeneities arise from multiple rooting systems here. The lateral HR plays a role in delivering some water to the center-most roots at a given soil depth. The presence of shallow understory roots drives overstory HR in the regions of high understory RWU (Fig. 9c and i). This HR-RWU interaction reduces the local minimum in soil water potential over longer periods (Fig. 9i) thereby sustaining understory transpiration through overstory-redistributed water. Model simulations reveal that redistribution by the overstory increases the HR fluxes located in the region occupied by understory roots (as percentage of total ecosystem HR, r_u) from 5.6% in the case of no competition to 11.2% and

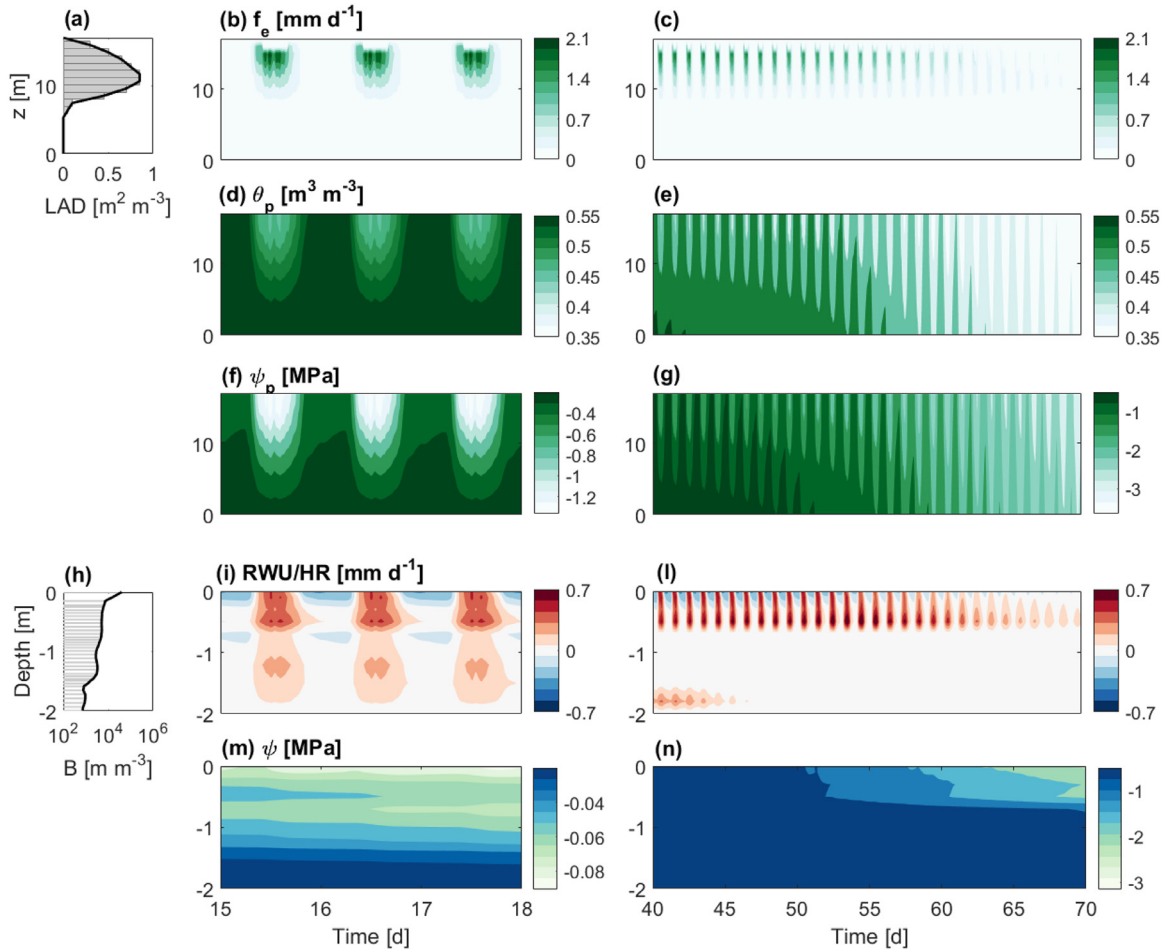


Fig. 4. 1D-1D Model results (overstory): modeled above (b–g) and below (i–n) ground water dynamics under well-watered (left) and water stress (right) conditions. Vertical distribution of (b–c) transpiration, (d–e) stem water content, (f–g) xylem potential, (i–l) total RWU/HR fluxes, and (m–n) soil water potential. LAD and root density profiles are respectively shown in (a) and (h).

22.8% in the case of water and light-water competition, respectively (see Fig. 9). Thus, competition for water acts to redirect the HR fluxes towards understory roots (Fig. 9b,e and h) while canopy shading enhances HR (Fig. 9c,f and i) by creating more favorable soil water potentials (i.e. less negative water potential and higher soil-root conductance) through reduced transpiration demand.

In this “model world”, competition for light and water may act to enhance resilience (i.e. sustained carbon dioxide and water fluxes) of both understory and overstory trees. This enhanced resilience occurs through two inter-related mechanisms: (i) a decreased evaporative demand (i.e. transpiration) due to shading that reduces carbon assimilation but sustains carbon/water uptake during prolonged periods of drought, (ii) localized vertical as well as lateral HR that sustains understory transpiration (Fig. 8). At the ecosystem scale, similar results are observed (Fig. 10a and b): competition for water accelerates the onset of water stress (Manoli et al., 2014) while light-water competition reduces carbon/water fluxes sustaining them over a longer period of time and for lower root-zone averaged soil water potentials, thus delaying stress conditions during prolonged drought.

Despite such complex interactions, the upscaled parameters of an equivalent RC model do not show substantial changes (Fig. 10). The magnitude of some parameters (namely equivalent stomatal conductance and tree capacitance) change among scenarios, but the bulk relations follow similar trends, indicating that such dynamics may still be captured by RC models by means of site-specific calibration. Hence, under specific conditions (e.g. when

the focus is on bulk quantities/fluxes and sufficient field data are available), more cost-effective approaches can be a valuable option. Since the porous media and RC approaches differ appreciably, the fact that the bulk parameters respond reasonably to resource fluctuations could not have been readily foreshadowed.

4.3. Sensitivity analysis

To assess the impact of different photosynthetic parameters and root density distributions (B) on simulated understory-overstory competition, a sensitivity analysis is performed (Fig. 11). In particular, understory transpiration subject to overstory shading (as in Fig. 7c and d) is simulated assuming a change of $\pm 20\%$ in the parameters $\lambda_{\max}$ and γ (Fig. 11a). Note that $\lambda_{\max}$ regulates stomatal functioning and γ controls photosynthesis in the case of light limitation (i.e. the condition the understory is subject to). Also, given that understory/overstory root niche separation can be beneficial to sustain RWU during water stress conditions Ivanov et al., the 3D-1D scenario illustrated in Fig. 9c is run considering different understory root distributions (Fig. 11b). To avoid any confounding effect, only the root density profile B is changed, while the total root biomass is preserved.

The results in Fig. 11a demonstrate that any increase in transpiration (e.g. due to a higher quantum yield or lower water use efficiency) speeds up the drying of the soil, anticipating the onset of water stress (Manoli et al., 2014). Similarly, a reduction of understory root biomass in the top soil (case B2 in Fig. 11b), limits

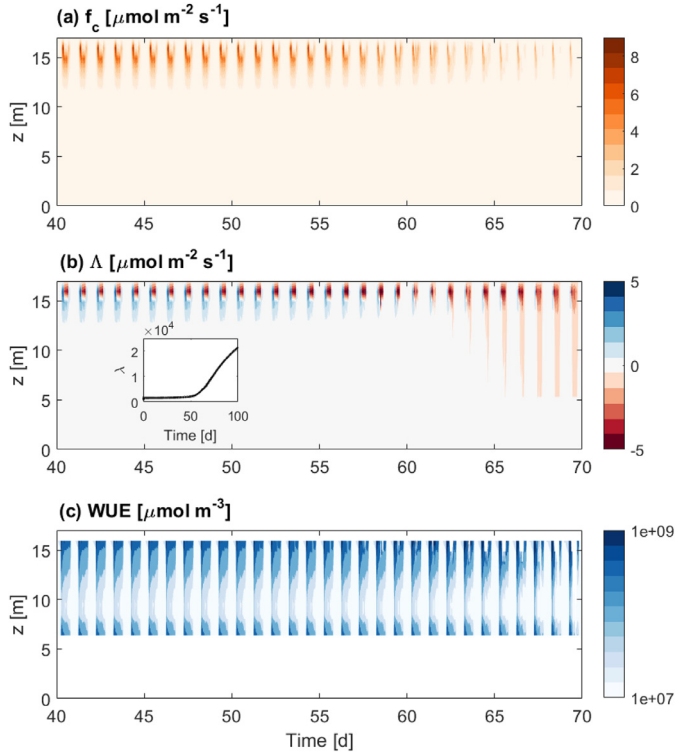


Fig. 5. 1D-1D Model results (overstory): Carbon uptake (a), objective function Λ regulating complete stomatal closure (b), and water use efficiency, WUE (c). According to the optimality hypothesis (Huang et al., 2017; Katul et al., 2010; Manzoni et al., 2011), when the water cost ($\lambda \cdot f_c$) exceeds the carbon gain (f_c), complete closure occurs (but cuticular losses are still allowed). The marginal water use efficiency λ is also shown in the inset of (b).

the access of the understory to overstory-redistributed water and accelerates the onset of water stress. Hence, investing carbon in the top soil (case B1) is shown to favor plant functioning when overstory-understory competition occurs. However, the impact of the root density distribution on plant transpiration is limited and a linear root profile (case B3) is also shown to maintain the same transpiration rate of case B1 (Fig. 11b). These results suggest that, to sustain transpiration, understory plants can either invest carbon in the top soil, i.e. where HR by the overstory is maximum, or to allocate more carbon in the deepest soil layers, i.e. where RWU by the overstory is minimum and the soil wetter (due to capillary rise). However, understanding if the co-existence of overstory and understory rooting systems is characterized by “optimum” root profiles that enhance plants resilience requires further investigations (also accounting for the 3D distribution of roots) and is left for future work. In general, the sensitivity analysis here demonstrates that different physiological parameters and root profiles can alter the timing of water stress but the overall dynamics are unchanged.

5. Limitations and perspectives

The focus here is on competition for water and light using a 1D canopy structure and light attenuation. In reality, branching architecture and LAD is three-dimensional, with direct consequences on photosynthesis and xylem hydrodynamics. For example, the occurrence of sunflecks (a pulse of direct sun beam that occurs over intervals of few minutes) has been shown to sustain understory photosynthesis appreciably (Naumburg et al., 2001). Such stochastic nature of light fluctuations over short durations and the ability of stomata and photosynthesis to adjust to them is a subject of active inquiry (Vico et al., 2011). Intermittent light availability, species-specific responses to sunflecks and stomatal opening and closing delays can be crucial in light-limited environment

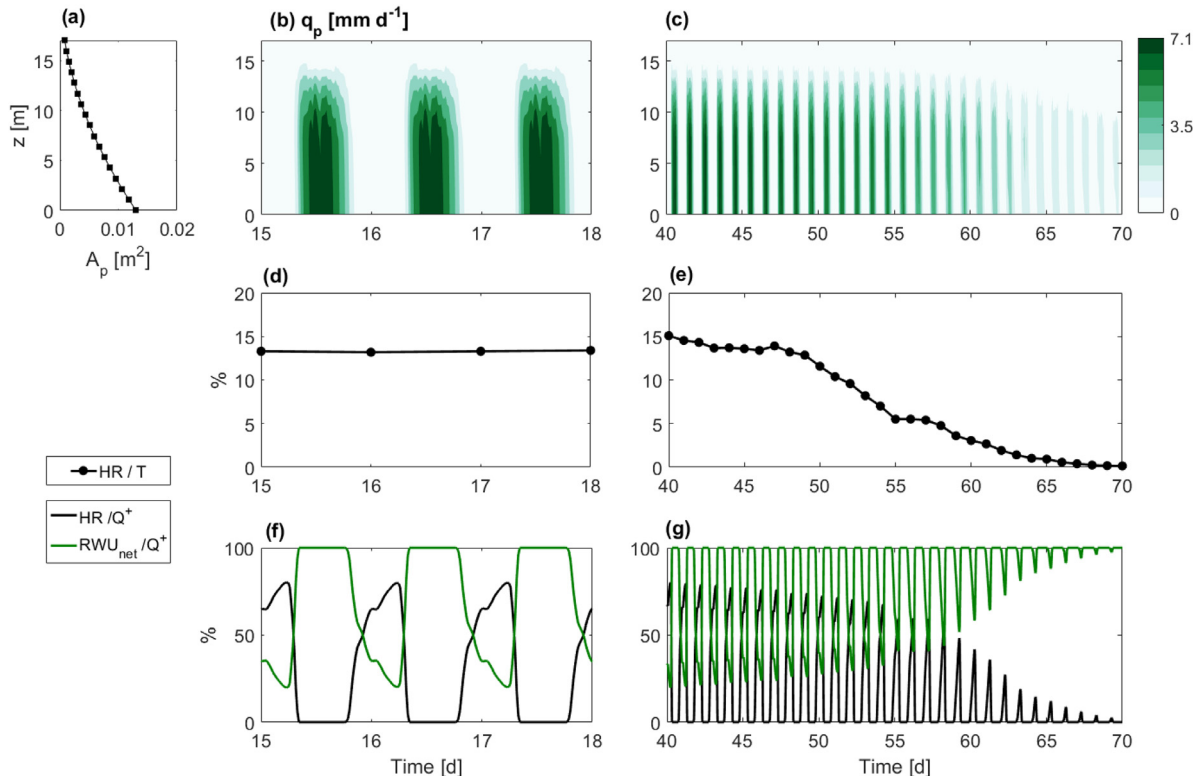


Fig. 6. 1D-1D Model results (overstory): vertical profile of xylem flux per unit leaf area (b-c), daily HR flux as a fraction of transpiration T , and partitioning of total RWU (Q^+) into RWU_{net} and HR (f-g) under well-watered (left) and water stress (right) conditions. The xylem cross section area, A_p , is also illustrated in (a) for reference.

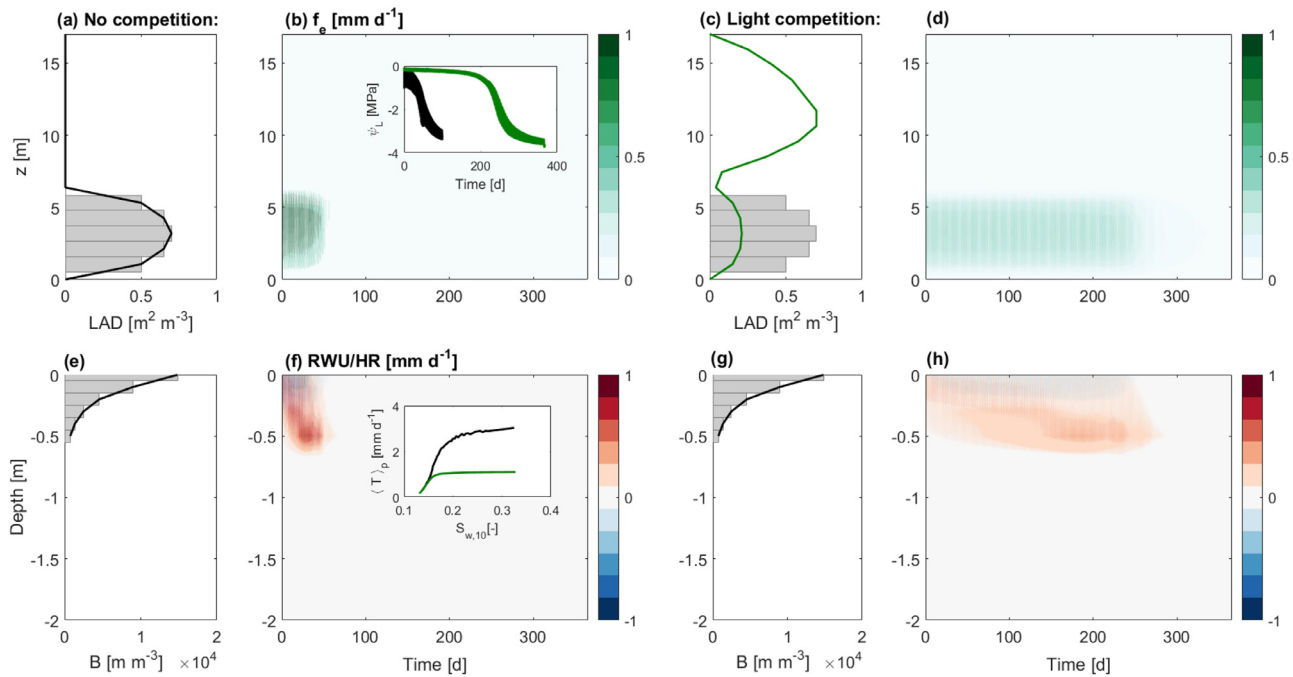


Fig. 7. 1D-1D Model results (understory): (b-d) transpiration and (f-h) RWU/HR fluxes with (right panels) and without (left panels) competition for light (overstory shading). The vertical distributions of (a,c) LAD_u (bars) and LAD_{ou} (black line in the case of no-competition, green line in the case of overstory shading), and (e-g) root density are also shown. Transpiration is calculated using LAD_u while light attenuation is calculated using LAD_{ou} . The inset in (b) shows the leaf water potential with (green) and without (black) light competition, while the inset in panel f shows transpiration as a function of soil saturation at 10 cm depth ($S_{w,10}$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Naumburg et al., 2001; Vico et al., 2011) and should be accounted for in future model developments.

In addition, a synthetic dry-down experiment and an idealized 3-trees ecosystem have been considered. Under such controlled conditions, tree-to-tree interactions are clearly disentangled as other confounding effects are neglected. However, the impact of climate variability and the role of different spatial configurations of trees on competition and long-term responses to drought should be further investigated. For example, the root uptake region of tall overstory trees may extend beyond the 1.5 m radius assumed here. Hence, our results are valid under the assumption of minor interactions among overstory trees. Increased root overlapping can enhance soil drying, altering the timing of the onset of water stress (Manoli et al., 2014). Moreover, the approach here assumes that atmospheric CO_2 , water vapor, air temperature, and air relative humidity are well mixed inside the canopy. This assumption is clearly violated in real forested canopies and attempts to correct for it require detailed models for turbulent flows within canopies. A number of studies have been conducted to assess the significance of the well-mixed assumption on sensible heat flux, water vapor, and CO_2 . It appears that when the boundary layer conductance is much larger than the stomatal conductance, the well mixed assumption is reasonable for modeling photosynthesis and leaf transpiration but not sensible heat (Juang et al., 2008). The latter also requires detailed accounting for leaf boundary layer conductance and leaf energy balance (Huang et al., 2015).

Also, the same parameters have been used here for both overstory and understory trees (i.e. young vs mature trees). This approach allows to disentangle light/water competition mechanisms but, in general, understory vegetation may have different physiological and xylem hydraulic characteristics developed to co-exist with overstory species (e.g. higher light-use efficiency). Such interspecies dynamics are likely not to impact the qualitative aspects of the conclusions (regarding the distribution of HR and the upscaling of fluxes), but they may have important ecological consequences

that go beyond the study scope (and more likely on longer time scales). It is also to be noted that the sign of Λ is sensitive to the choice of the photosynthetic parameters and the leaf water potential, hence different cost-gain trade-off can be observed for different plant species or characteristics (e.g. LAD).

The model for photosynthesis assumes that the mesophyll conductance is much larger than the stomatal conductance. A large number of studies on droughts and salt-stress conditions seem to indicate that mesophyll conductance may be significant and should be accounted for (Sun et al., 2014; Volpe et al., 2011). The development of models that predict mesophyll conductance from leaf water status (and osmotic potential) is currently lacking due to data availability but is ripe for progress.

Regarding plant hydraulics, a “big root” model is here employed (Siqueira et al., 2008) and radial flow close to the rootlets (known to be important to capture the RWU diurnal cycle (Tuzet et al., 2003)) as well as flow within the rooting system are neglected. Modeling the coupled soil-root water transport may be needed in certain situations as the root hydraulic conductivity can significantly influence the HR fluxes (Amenu and Kumar, 2007; Quijano and Kumar, 2015). However, energy losses are much smaller within roots than in the soil and across the root-soil interface (Siqueira et al., 2008), thus justifying the working assumptions of a lumped soil-to-root conductance. Similarly, a macroscopic representation of the plant hydraulic conductivity K_p that lumps energy losses along the xylem length and across torus-margo or end-member pits has been considered (Sperry, 2003). These considerations suggest that resistances along end-pits connecting xylem tubes are proportional to frictional losses along the lumen tube length (Hacke et al., 2006; Hunt and Manzoni, 2015; Wheeler et al., 2005). Indirect evidence for this coordination between frictional losses along the lumen length and connecting pits is the emergence of a scaling relation between lumen length L_L and radius r_L given by $r_L \sim L_L^{2/3}$. If such proportionality holds, the energy losses across “joints” (i.e. torus-margo or end-member pits) can be re-

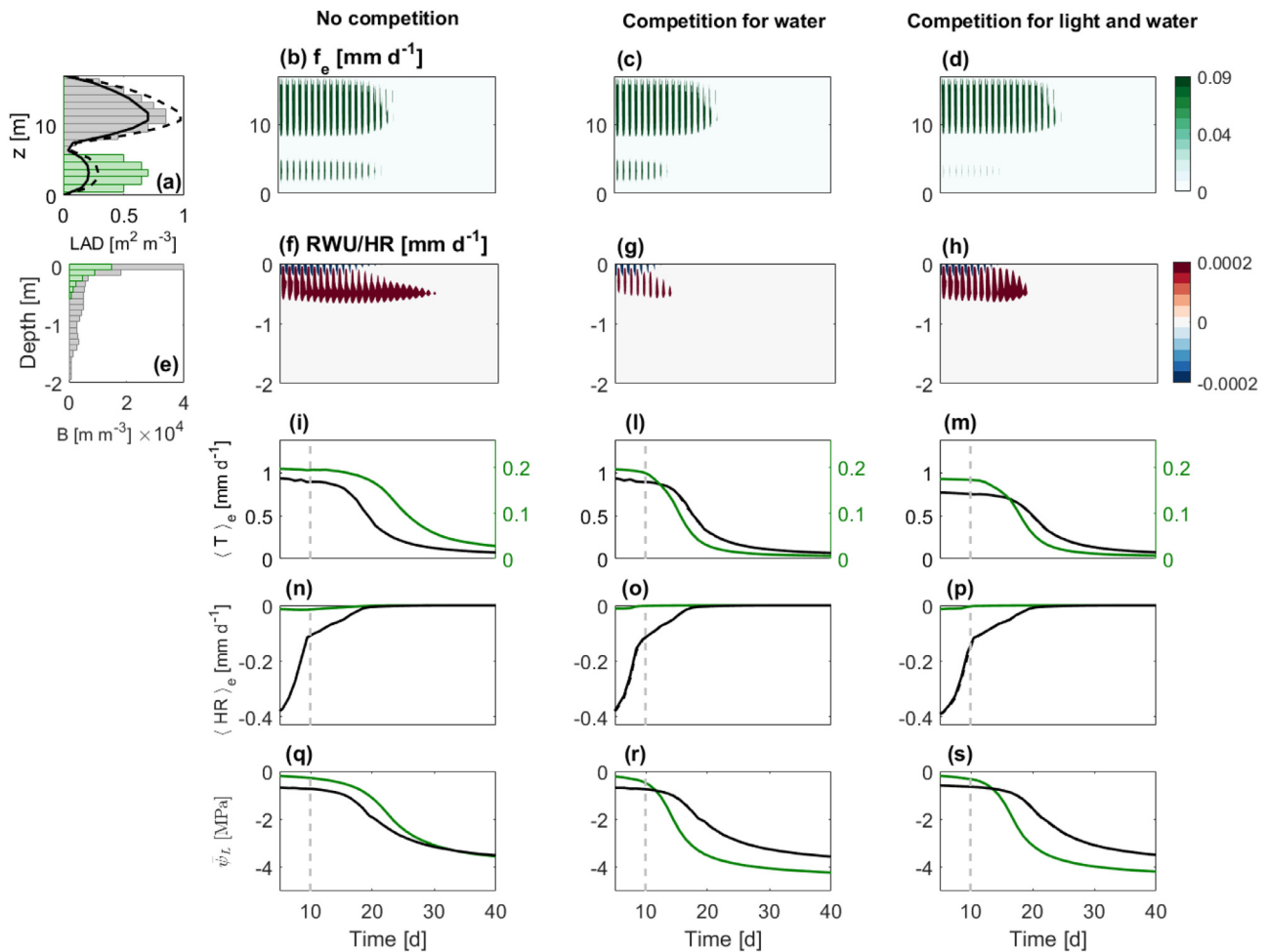


Fig. 8. 3D-1D Model results: modeled transpiration (b-d, i-m), RWU/HR fluxes (f-h, n-p) and leaf water potential (q-s) in the case of no competition (left panels), competition for water (central panels), and competition for both light and water (right panels). The dashed gray line in panels i-s indicates the time illustrated in Fig. 9. The vertical distribution of LAD and root density of understory (green bars) and overstory (gray bars) trees is illustrated in (a) and (e), respectively. The LAD_{uo} profiles used in the case of no competition/competition for water (solid black line) and competition for light and water (dashed black line) are also shown in (a). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

placed by an equivalent friction factor in a Hagen-Poiseuille equation proportional to the one used for the tube length. This means that the flow-rate is proportional to the pressure gradient and the overall xylem resistivity to water transport may still be approximated by a Darcy representation such as the one employed here, even though the “junction” losses are not a frictional loss along a tube length. A large corpus of data across many species (including deciduous and evergreen angiosperms and gymnosperms) suggest that $r_L \sim L_L^{2/3}$ reasonably holds except for lianas (Hunt and Manzoni, 2015).

Due to the 1D approximation, other tree-scale processes have been neglected (e.g. canopy interception, preferential flow, heterogeneous shading). The compound dynamics would require a detailed 3D description of the canopy structure and is left for future work. However, recent advances in modeling 3D canopies (Beyer et al., 2014) and remote sensing (now providing reliable estimates of canopy profile, LAI, and leaf angle distributions (Zhao et al., 2015)) may guide future work to include realistic representation of above ground processes. Notwithstanding these limitations, the 1D above-ground representation is particularly suitable to maintain low computational costs and is consistent with the beam representation of trees employed in state-of-the-science Large Eddy Simulations (LES) (Dupont et al., 2015; Pivato et al., 2014). At present, LES are not fully integrated with soil-

plant hydrodynamics, but such a coupling may provide new foresight on the role of vegetation in regulating canopy microclimate, boundary layer growth, and small scale (~ 1 -10 m grid size) land-atmosphere interactions.

6. Conclusions and future research directions

A novel approach to represent soil-plant interactions for the purpose of exploring tree-to-tree competition for light and water among multiple trees has been presented. The main features of the proposed model include coupling a 3D description of soil moisture dynamics and RWU mechanisms to a 1D representation of above ground processes (stem flow and storage, photosynthesis, transpiration). This representation accounts for the appreciable vertical light attenuation within the canopy (modeled as a homogeneous isotropic medium). At the plant scale, simulation results reveal complex soil-plant dynamics characterized by: (i) a vertical distribution of transpiration regulated by light attenuation and carbon gain-cost trade-offs, (ii) light attenuation reducing the evaporative demand and enhancing the resilience of competing understory trees to drought, and (iii) a competition between understory and overstory roots so as to generate heterogeneous RWU/HR fluxes that then support sustained understory transpiration and carbon gain. In particular, model simulations suggest that HR by the overstory is comparable with understory transpiration and competition

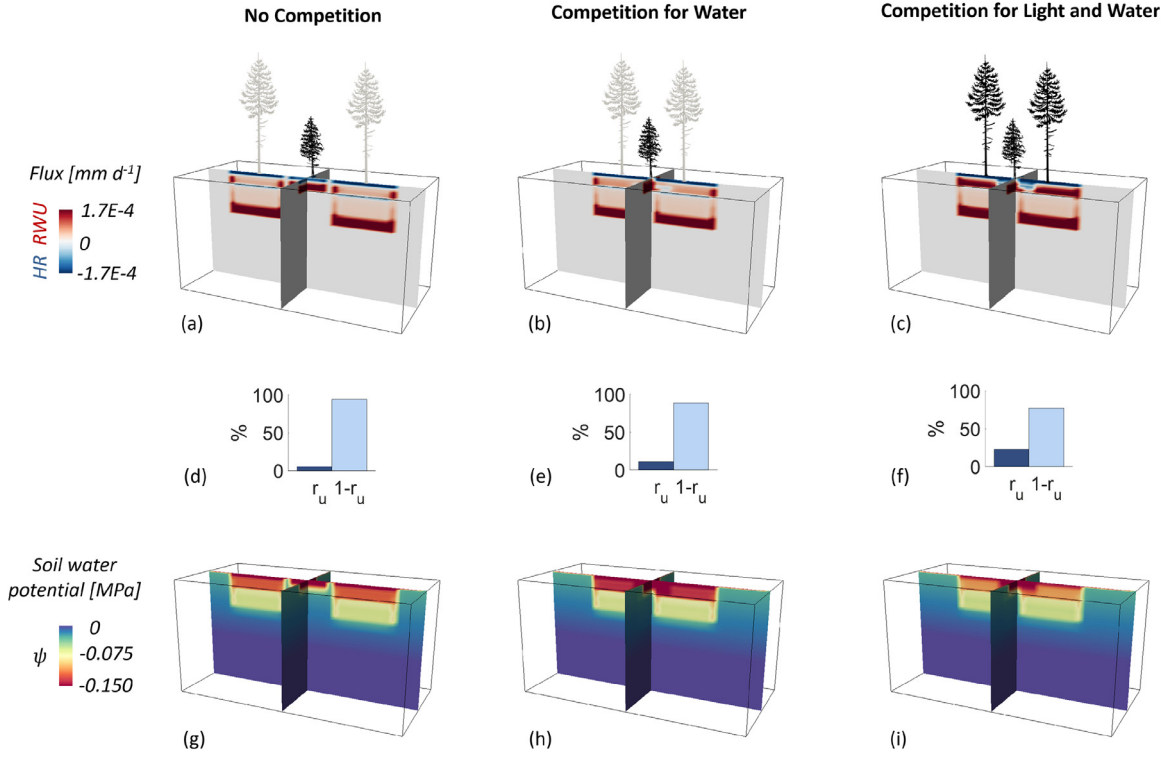


Fig. 9. 3D-1D Model results: spatial distribution of (a-c) RWU, HR fluxes and (g-i) soil water potential at 10 PM (day 10, see dashed line in Fig. 8) in the case of (a,g) no competition, (b,h) competition for water and (c,i) competition for both light and water. The percentage of HR located in the region occupied by understory roots ($r_u = \langle HR \rangle_{urz} / \langle HR \rangle_e$, being $\langle HR \rangle_{urz}$ the flux occurring in the understory root zone and $\langle HR \rangle_e$ the total ecosystem flux) is also shown (d-f).

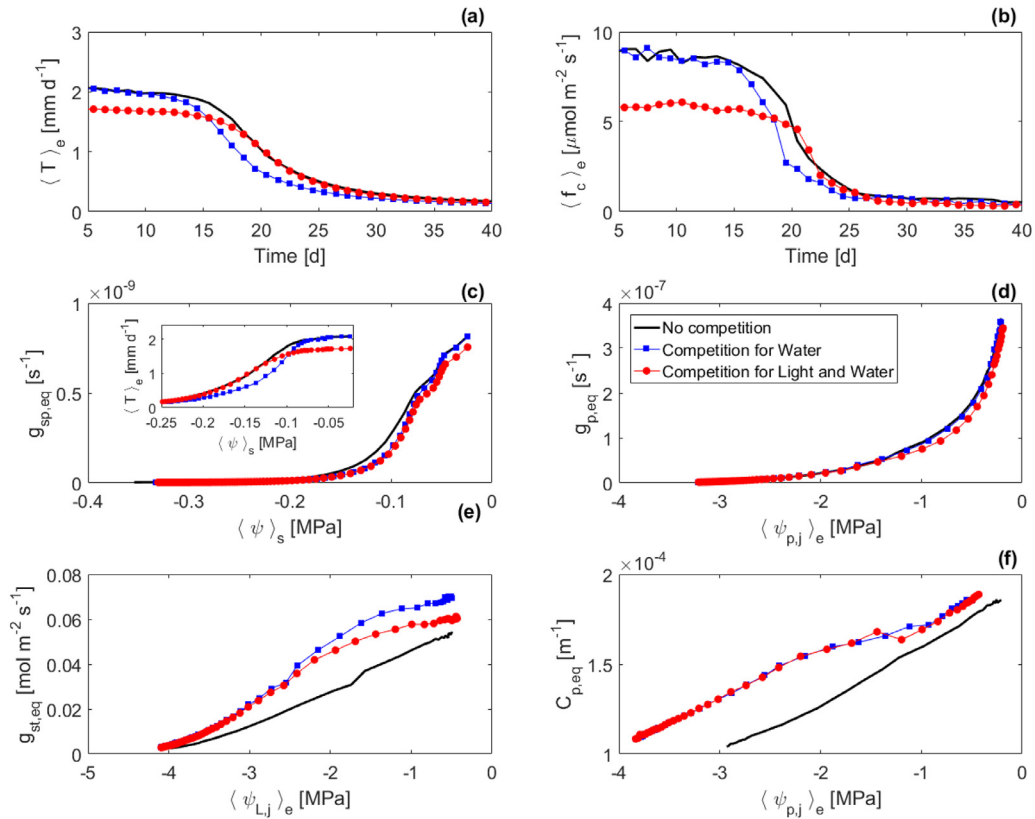


Fig. 10. 3D-1D Model results: ecosystem (a) water and (b) carbon fluxes, (c) soil-plant, (d) xylem, and (e) stomatal conductances and (f) capacitance of the equivalent RC system fitted to the full model (see Fig. 3). Since $C_{p,eq}$ is ill defined when $d\langle \psi_{p,j} \rangle_e$ is near zero (see Eq. (20) and Bohrer et al., 2005), it is only calculated for $d\langle \psi_{p,j} \rangle_e > 2$. Ecosystem transpiration as a function of average soil water potential is also shown in the inset of (c).

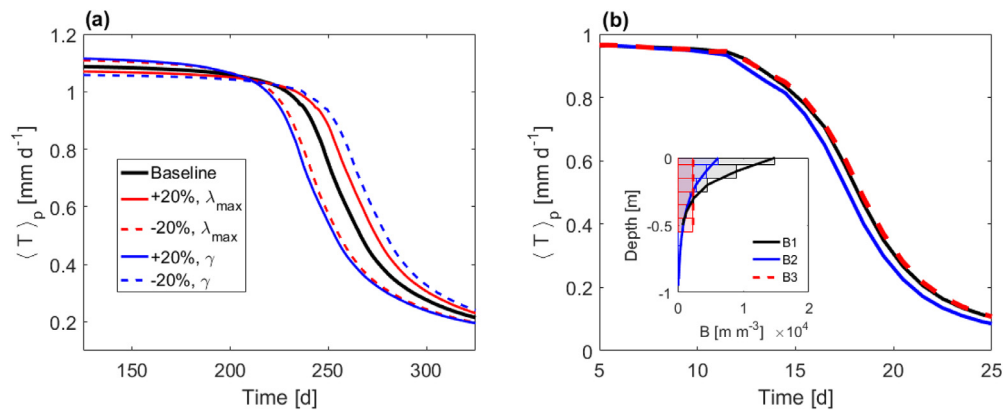


Fig. 11. Sensitivity analysis: changes in understory transpiration due to (a) changes ($\pm 20\%$) in the physiological parameters $\lambda_{\max}$ and γ (1D–1D simulations with light competition as in Fig. 7d) and (b) different understory root density distributions B1–B3 (3D–1D simulations with light and water competition as in Fig. 8d,m). The baseline scenario in (a) and the simulations illustrated in (b) were run using the model parameters listed in Table 2. The root profiles B1–B3 are shown in the inset of panel b. Note that the root density profile B1 is the same illustrated in Figs. 7e,g and 8e.

for soil water increases the availability of overstory-redistributed water for understory RWU from 6 to 23%. At the ecosystem scale, such complex dynamics can be lumped into bulk relations, suggesting that “zeroth-order” spatial models may capture upscaled effects but rigorously deriving the effective parameters for such reduced model remains a subject for future inquiry.

At the individual tree scale, future research should focus on modeling carbohydrate transport in the phloem and access to soil nutrients for a more realistic description of carbon allocation and growth processes (Fatichi et al., 2014). An improved description of land-surface characteristics (i.e. complex topography, surface-subsurface water interactions) and land-atmosphere coupling is also needed to gain insight into the role of plants in regulating canopy micro-climate, boundary layer dynamics (e.g. by coupled SPA–LES simulations) and possible self-regulation mechanisms within the hydrologic cycle (Manoli et al., 2016).

Acknowledgments

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